

Future Interns - Cyber Security Task 2

SECURITY ALERT MONITORING & INCIDENT RESPONSE

Task:

- Monitor simulated security alerts using a SIEM tool, identify suspicious activities, classify incidents, and draft an incident response report

Key Features to Include

- Ability to identify 3–5 suspicious alerts from logs
- Incident classification by priority (High, Medium, Low)
- Detailed incident response report with timeline, impact, and remediation suggestions
- A summary dashboard screenshot from the SIEM tool
- (Optional) A communication email template reporting the incident to management

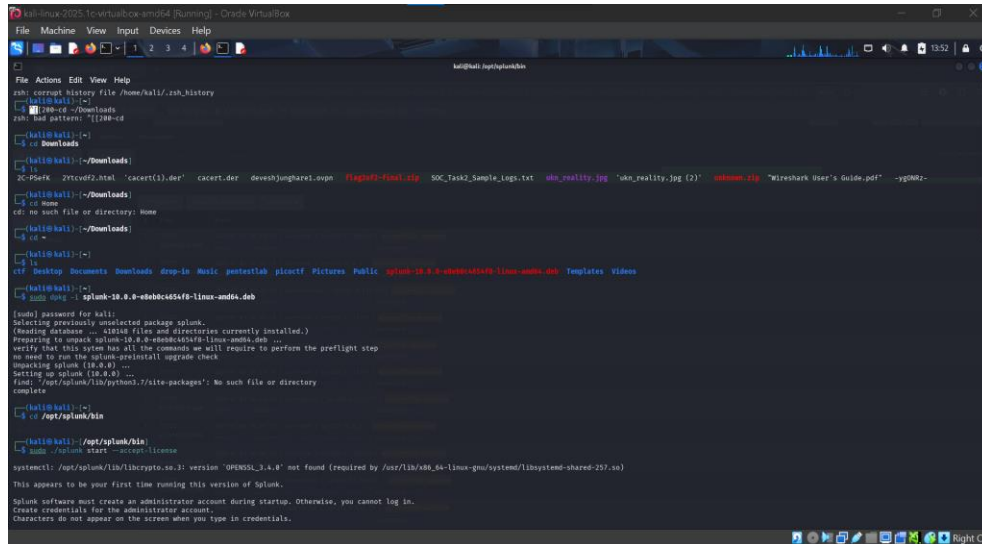
Tools:

- Elastic Stack (ELK) – Open-source SIEM for ingesting and analyzing logs ([Elastic.co](https://www.elastic.co))
- Splunk Free Trial – Powerful SIEM platform ([Splunk.com](https://splunk.com))

Set up Splunk Free Trial (v10.0.0):

1. Installation:

- Installed Splunk Enterprise 10.0.0 on a Windows machine.
- Completed the initial setup and accepted the license agreement
- Created Username and Password
- Generates an Local host ip website for Dashboard



```
kali@kali:~/Downloads$ dpkg --get-selections | grep splunk
splunk:install
kali@kali:~/Downloads$ dpkg --get-architecture
amd64
kali@kali:~/Downloads$ dpkg --get-installer splunk
./splunk-8.8-ubuntu64bit-linux-amd64.deb
kali@kali:~/Downloads$ dpkg --get-filename splunk
./splunk-8.8-ubuntu64bit-linux-amd64.deb
kali@kali:~/Downloads$ dpkg --get-selections | grep splunk
splunk:install
```

2. Adding Data:

- Navigated to the “Add Data” section on the Splunk dashboard.
- Uploaded the sample logs provided by Future Interns.
- On the “Set Source Type” page, selected the default type for simplicity.
- Reviewed the input settings and confirmed all configurations
- Using the search bar, I was able to apply SQL-like queries to extract specific parts of the logs and record them into a table. For example: `index="main" sourcetype="future interns log analysis" action="login failed"`.

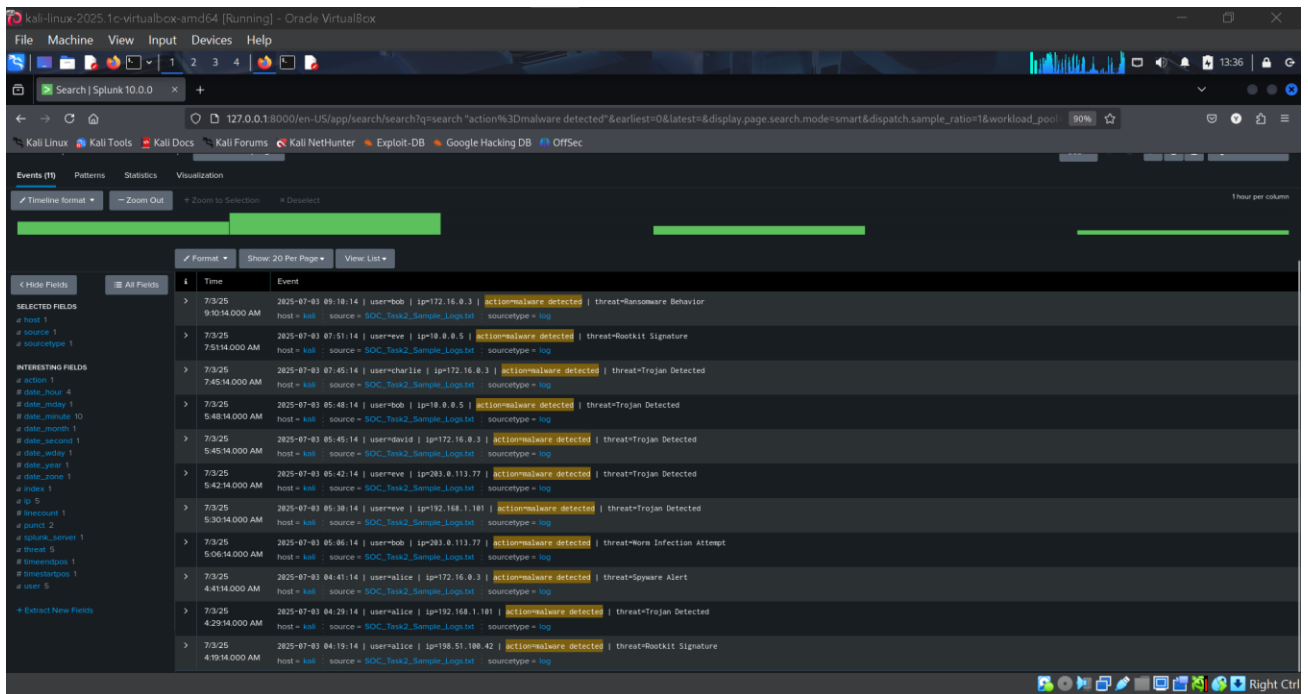
3. Searching and Filtering Logs:

- Used Splunk’s search bar to run queries for specific events, e.g.,
- `index="main" sourcetype="future interns log analysis" action="login failed"`
- This returned only failed login events, saving time by avoiding manual scrolling through all logs.
- Similarly, other filters can be applied to quickly extract malware alerts, file access events, or connection attempts.

Log Analysis:

1. Malware Alert:

- A malware alert in a log file is a notification generated by a security system indicating that a malicious activity or file has been detected on the system or network. It typically includes the type of malware (e.g., Trojan, worm, spyware), affected IP or device, severity level (high, medium, low), timestamp, and sometimes the action taken (blocked, quarantined, or removed).



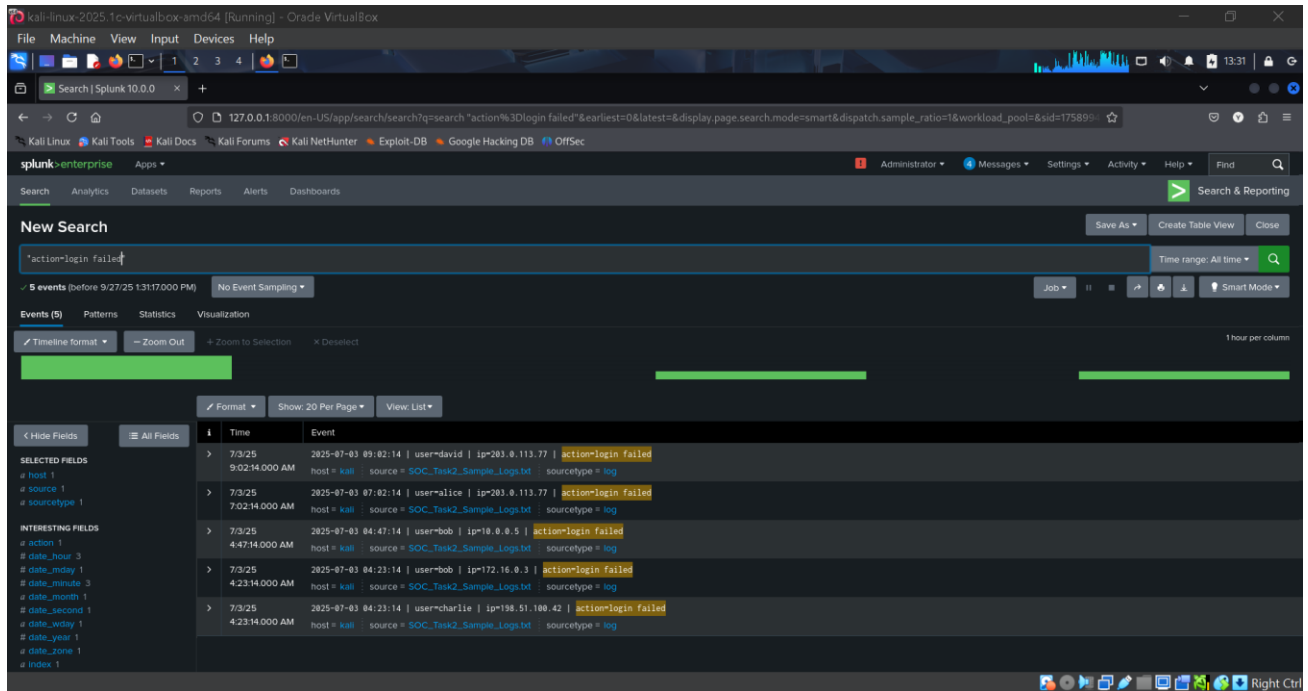
- This Screenshot highlights multiple malware detections affecting several users and significantly impacting the system. Malware types observed include Trojan, Worms, Spyware, among others. Notably, the IP address 172.16.0.3 experienced multiple malware detections, suggesting it is a heavily compromised endpoint.
- Threat (High):
The system detected multiple malware infections, particularly targeting the user associated with 172.16.0.3, indicating a lack of cybersecurity awareness and potential vulnerability in user practices.
- Impact:
If left unaddressed, these malware infections could perform malicious activities such as:
 1. Deleting user files
 2. Encrypting sensitive data
 3. Exfiltrating confidential information

Recommended Next Steps:

1. Cybersecurity Awareness Training: Educate users on basic cybersecurity practices, including recognizing potential threats like Trojans.
2. Anti-Malware Solutions: Install and maintain reliable antivirus or endpoint protection software to detect and prevent malware infections.
3. Patch Management: Identify and remediate vulnerabilities promptly to prevent exploitation by malware.

2. Failed login

- A **FAILED LOGIN** entry in a log file indicates that someone tried to access a system or account but was **unsuccessful**. These entries usually include: **timestamp**, **username**, **source IP**, and sometimes **the reason** for failure.



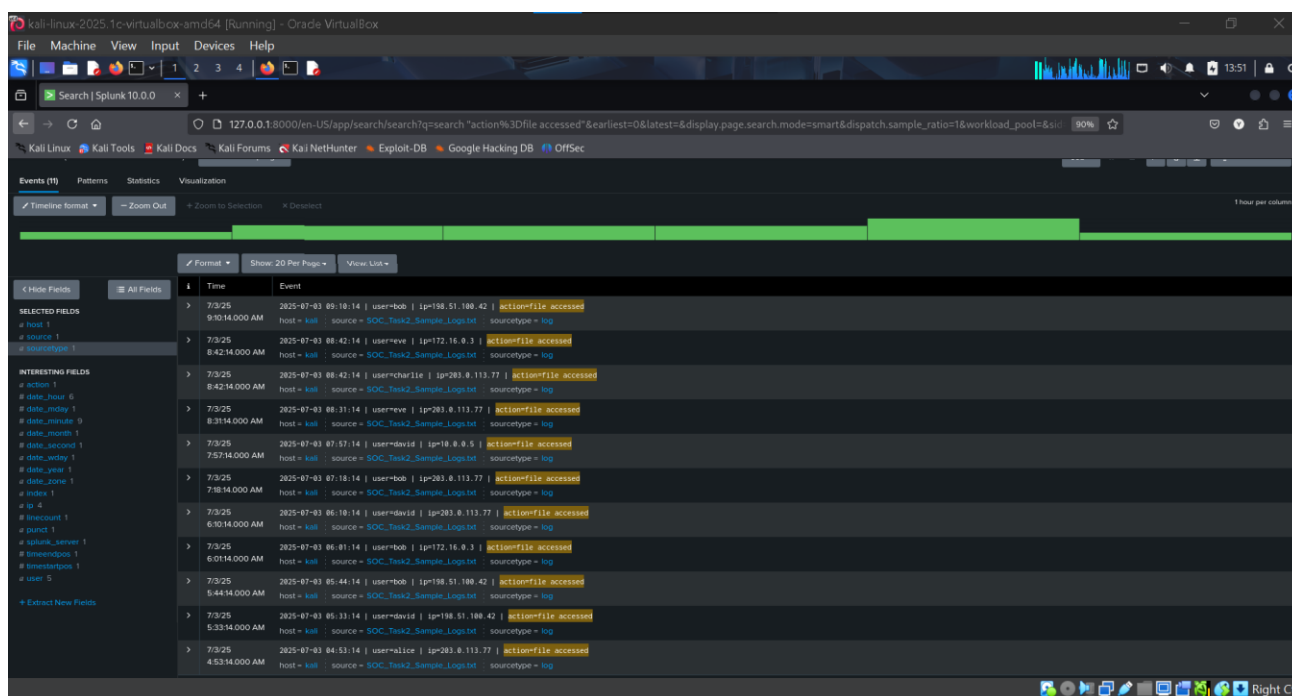
- As observed, two accounts attempted to log in using the same IP address. User Alice tried to log in but failed, and two minutes later, user David attempted to log in from the same IP address, also unsuccessfully. Additionally, multiple failed login attempts were recorded for users David, Alice, Bob, and Charlie, indicating a brute-force attack.
- Threat (Low): Repeated login attempts, especially from IP address 203.0.113.77, targeting multiple user accounts without success, indicate suspicious activity.
- Impact: If left unmitigated, these attempts could lead to account compromise, granting unauthorized access to sensitive information.

Recommended Next Steps:

- Firewall Protection: Block suspicious IP addresses, such as 203.0.113.77.
- Strong Password Policies: Encourage users to use complex, unique passwords.
- Account Lockout Policy: Configure accounts to lock after a specified number of failed attempts (e.g., 3) to prevent brute-force attacks.

3. Files accessed

- A FILE ACCESSED entry in a log file indicates that a user or process has opened, read, or modified a file on the system. These logs usually include the timestamp, username or process, file path, and action performed.



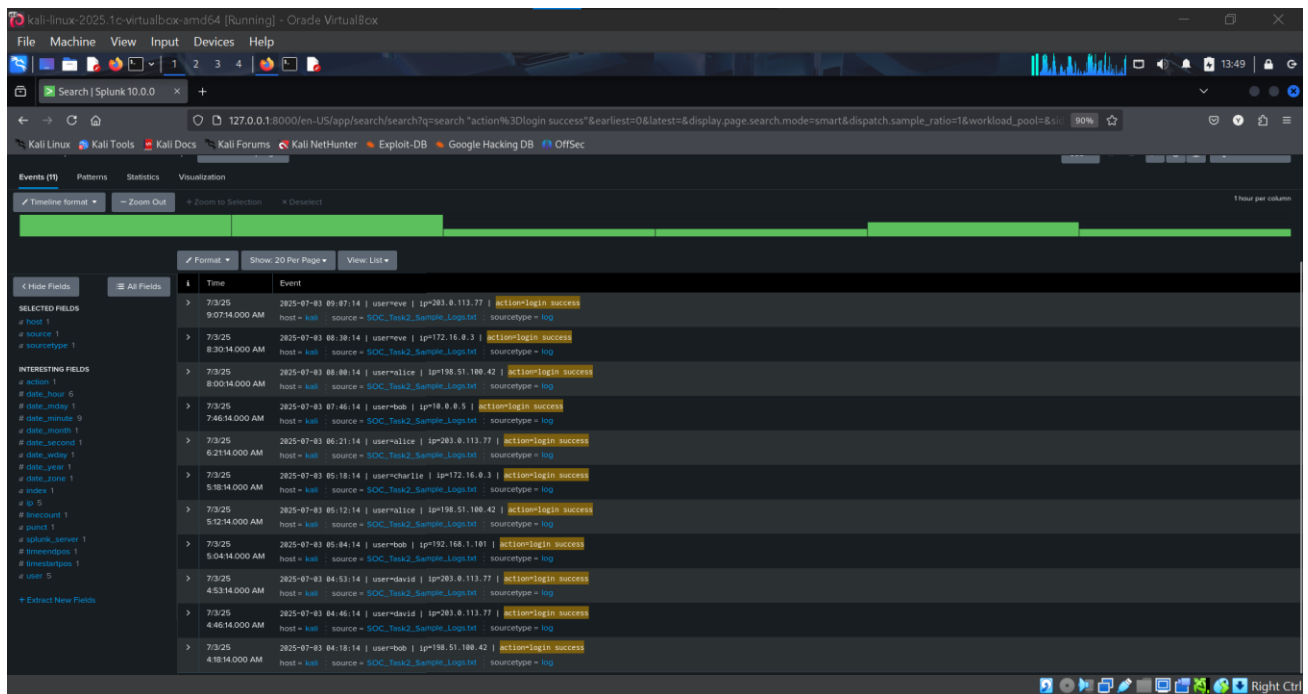
- This screenshot shows that a user was able to access a file; however, the previously identified malicious IP address (203.0.113.77) appears multiple times in this table and successfully accessed a file. This indicates that the files could potentially be exploited for malicious purposes such as ransomware attacks or file deletion.
- Threat (High):
Access to files by a malicious IP address poses a significant risk. If not addressed, the accessed files could be used for harmful activities, compromising the system's integrity.
- Impact:
There is a potential for data theft or unauthorized viewing of sensitive information, which could have serious consequences for the organization.

Recommended Next Steps:

- Firewall Protection: Block any malicious IP addresses and restrict access to only authorized users.
- Data Monitoring: Check if any sensitive data has been exposed and take immediate action, such as closing compromised accounts, to prevent further damage.

4. Success Login

- A **SUCCESSFUL LOGIN** entry in a log file indicates that a user or process has successfully authenticated and gained access to a system or account. These logs typically include the timestamp, username, source IP, and sometimes the session ID or method of login.



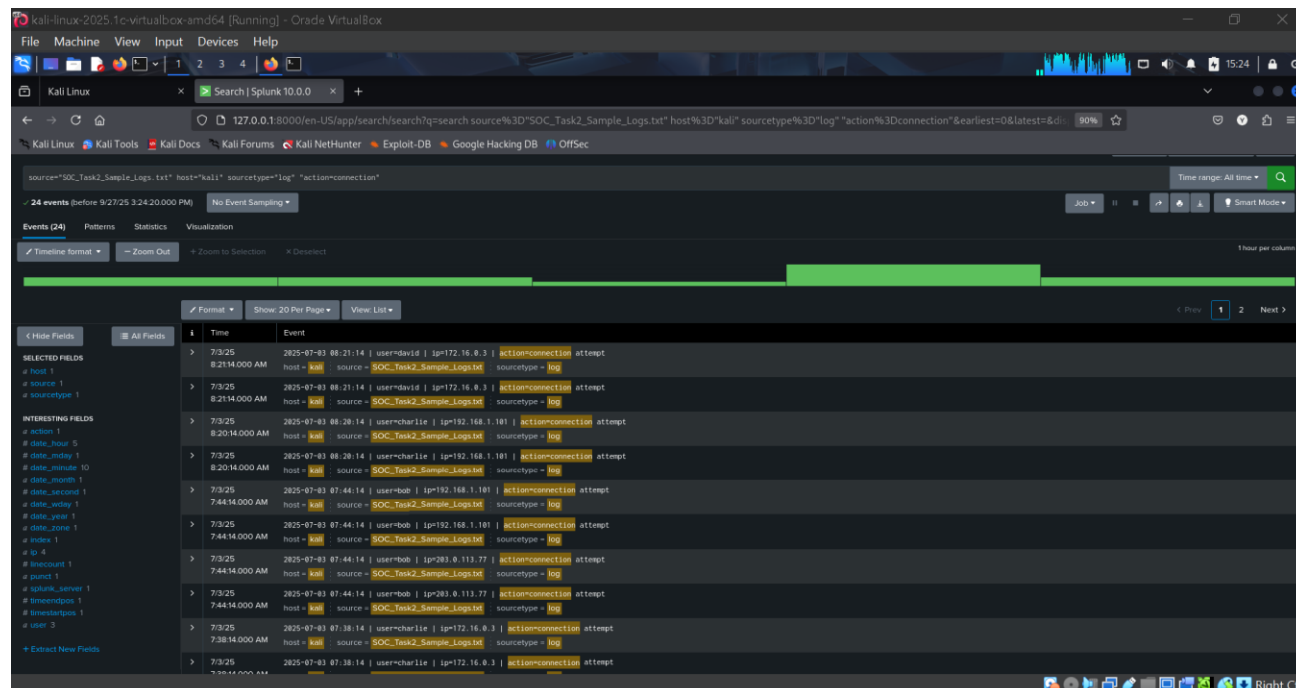
- This Screenshot records **successful logins**, indicating that an attacker may have gained **authorized access** after multiple failed login attempts. The suspicious IP address **203.0.113.77** appears frequently during the brute-force attempts, along with another IP address, **198.51.100.42**. This pattern demonstrates that the attacker was eventually able to bypass security controls, highlighting a vulnerability in the system's ability to handle brute-force attacks.
- Threat (High):**
The attacker may have successfully gained unauthorized authorized access to the system after repeated attempts, representing a serious security risk.
- Impact:**
This could lead to exposure of sensitive data, potential ransomware attacks, and significant legal and reputational consequences for the organization due to insufficient security controls.

Recommended Next Steps:

- Implement Multi-Factor Authentication (MFA):** Even if an attacker gains access to credentials, MFA adds an additional verification layer to prevent unauthorized access.
- Introduce CAPTCHA or Human Verification:** Adding a CAPTCHA-style puzzle during login can help prevent automated brute-force attacks by distinguishing human users from automated login attempts.

5. Connection Attempt

- A CONNECTION ATTEMPT entry in a log file indicates that a system or device tried to establish a network connection, whether successful or not. These logs usually include the timestamp, source IP, destination IP/port, protocol, and result (success/failure).



- Threat (Medium):**
This log shows connection attempts from both internal and external IP addresses. The recurrence of these IP addresses, as seen in previous tables, indicates potential lateral movement attempts within the network.
- Impact:**
If not mitigated, attackers could propagate across the system, increasing the risk of further compromise and enabling additional attacks on critical assets.

Recommended Next Steps:

- Block External Malicious IPs:** Prevent known malicious external IP addresses from connecting to the network.
- Network Segmentation:** Isolate critical systems and restrict internal communication paths to limit lateral movement.
- Monitor Internal Traffic:** Continuously observe internal connections for unusual or unauthorized activity to detect suspicious behavior early.

Learn how SOC teams track and manage threats using dashboards and playbook:

- Using the Splunk dashboard, I was able to visualize and filter security alert events efficiently. For instance, the screenshot provided below shows a table of failed login attempts, including details such as username, suspicious IP address, and timestamp. By transforming raw log data into structured tables, dashboards make it significantly easier to identify and analyze suspicious events without scrolling through large volumes of unorganized data.
- Additionally, SOC teams utilize playbooks, which are step-by-step guides that outline how to respond to different types of incidents, such as brute-force attacks, Trojans, spyware, and other malware. Playbooks help ensure that analysts:
 - Follow a consistent and standardized process for incident response
 - Reduce human errors during analysis and remediation
 - Respond quickly and effectively to emerging threats

By combining dashboard visualizations with playbooks, SOC teams can monitor, detect, and respond to security incidents in a structured and efficient manner.

